Chapter 44 — Application Lifecycle Management

> **Part VI — Lifecycle, Background & Platform** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 6 (UI state/ViewModel), Ch 8 (process death), Ch 5 (Hilt/ Application), Ch 14/41 (background batching, flush on background). Foundational for Ch 45 (startup), Ch 46/47 (services/Doze), Ch 49 (push).
>
> **Reference surface in StreamCart:** knowing when the app is foreground vs background, initializing/tearing down app-wide resources at the right time, and doing lifecycle-correct work (flush analytics on background, refresh on foreground).

44.1 Problem Definition and Real-World Context

Android apps don't have a simple "running/stopped" model — there's a **process**, an **Application**, **Activities/Fragments** (or Compose destinations), and the whole app moving between **foreground and background**, each with its own lifecycle. Code that ignores these boundaries leaks, wastes battery, crashes, or misbehaves:

- **Work tied to the wrong lifecycle.** A network poll or location request started in an Activity keeps running after the user leaves (battery/leak), or a coroutine tied to a destroyed scope crashes.
- **No foreground/background awareness.** The app can't tell when it's backgrounded to flush analytics (Ch 14), pause expensive work, or refresh on return — so it polls while hidden or shows stale data on resume.
- **Initializing in the wrong place.** Heavy work in `Application.onCreate` slows every launch (Ch 45); per-Activity re-init duplicates work.
- **Leaks from lifecycle misuse.** Observing without lifecycle awareness, holding Activity references, or not unregistering callbacks leaks the Activity/memory.
- **Process-lifetime assumptions.** Treating the Application as permanent misses that the OS kills the process (Ch 8) and that `Application.onCreate` runs on every cold start.

The architecture: understand the **lifecycle hierarchy** (process → Application → Activity/Fragment/Compose), use **lifecycle-aware components** (`LifecycleOwner` , `repeatOnLifecycle` , `viewModelScope` , Ch 6) so work is scoped to the right lifetime, detect **app-wide foreground/background** via `ProcessLifecycleOwner` , and place initialization/teardown correctly. This chapter is about **doing the right work at the right lifecycle boundary** without leaks or waste.

Three practical use cases

1. **Foreground/background work (StreamCart).** Flush batched analytics (Ch 14) and pause real-time connections (Ch 63) when the app backgrounds; refresh/reconnect on foreground.
2. **Lifecycle-scoped work (all).** A screen's data collection stops when it's not visible (`repeatOnLifecycle` , Ch 6); a long task survives navigation via the ViewModel/WorkManager (Ch 30).

3. **App-wide init (all).** Initialize crash reporting/DI/logging once in `Application` (lightly, Ch 45), and app-wide observers via `ProcessLifecycleOwner`.

Why this is architectural

Lifecycle management defines **what runs when** (process/app/screen scopes), how work is **bounded and cancelled** (no leaks/waste), and how the app reacts to **foreground/background** — cross-cutting properties that determine correctness, battery, and memory across every feature.

44.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Scope work to the correct lifecycle (process/app/screen); cancel when the scope ends.
F2	Detect app-wide foreground/background transitions.
F3	Do lifecycle-correct reactions (flush/pause on background; refresh/reconnect on foreground).
F4	Initialize app-wide resources once, in the right place (lightly).
F5	Avoid leaks (lifecycle-aware observation; no retained Activity refs).
F6	Survive config change (ViewModel) and process death (Ch 8).

Non-functional requirements

Attribute	Policy
Correctness	Work bounded to its lifecycle; no crashes on destroyed scopes.
Battery	No background work while hidden (unless intended).
Memory	No lifecycle leaks (LeakCanary, Ch 89).
Performance	Light `Application.onCreate` (Ch 45).
Maintainability	Lifecycle-aware components, not manual callbacks.

Scope

- **In scope:** lifecycle hierarchy (process/Application/Activity/Fragment/Compose), `LifecycleOwner` / `repeatOnLifecycle` / `viewModelScope`, `ProcessLifecycleOwner` (app foreground/background), app-wide init/teardown, leak avoidance, lifecycle in Compose.
- **Out of scope:** startup optimization (Ch 45), services/Doze (Ch 46/47), process death/SavedState (Ch 8), UI state (Ch 6).

Assumptions & constraints

- Jetpack Lifecycle (`androidx.lifecycle`), Compose + `lifecycle-runtime-compose` , Hilt. `minSdk 24` . Process is ephemeral (Ch 8); `Application` recreated on each cold start.

44.3 System Design Fundamentals

- **The lifecycle hierarchy (nested lifetimes):**
  - **Process:** the OS process. Created on cold start, killed anytime in background (Ch 8). `Application.onCreate` runs once per process.
  - **Application:** the app-global object (`@HiltAndroidApp`), lives as long as the process. Place app-wide, cheap, one-time init here (DI, crash reporting, logging).
  - **Activity / Fragment / Compose destination:** UI lifecycles (`onCreate` / `onStart` / `onResume` / `onPause` / `onStop` / `onDestroy`), recreated on config change. Screen-scoped work lives here (but state lives in the ViewModel, Ch 6).
  - **App foreground/background:** the *whole app* moving between visible and hidden — observed via `ProcessLifecycleOwner` .
- **Lifecycle-aware components (the modern model).** Instead of manual `onStart/onStop` plumbing, use **`LifecycleOwner`** + **`LifecycleObserver`** / **`repeatOnLifecycle(STARTED)`** / **`lifecycleScope`** / **`viewModelScope`** (Ch 6) so work **automatically starts/stops/cancels** with the lifecycle. This is what prevents leaks and wasted work — e.g., `repeatOnLifecycle(STARTED)` collects a Flow only while the screen is visible.
- **`ProcessLifecycleOwner` (app-wide foreground/background).** A single `LifecycleOwner` for the **whole app**: `ON_START` when the app comes to the foreground, `ON_STOP` when it goes to the background (after the last Activity stops). Use it for app-level reactions: flush analytics (Ch 14), pause/resume WebSockets (Ch 63), pause expensive polling. **Don't** infer background from a single Activity's `onStop` (config changes / multi-Activity flows mislead).
- **Scope selection = correctness + battery.** Match work to the lifetime it should have: UI-only work → screen scope (stops when invisible); work that must survive navigation/config → ViewModel (Ch 6); work that must survive the UI/process → WorkManager (Ch 30). Starting long work in a screen scope is the classic bug.
- **Init placement.** App-wide, cheap, needed-early init → `Application.onCreate` (keep it light, Ch 45) or **App Startup** (Ch 45). Per-feature init → lazily when the feature is used (DI, Ch 5). Avoid heavy work in `Application.onCreate` — it runs on every cold start.
- **Compose lifecycle.** Composables observe the lifecycle via `collectAsStateWithLifecycle` and `LocalLifecycleOwner` ; `DisposableEffect` / `LaunchedEffect` tie work to composition + lifecycle (Ch 6/7).

Core insight: **scope every piece of work to the correct lifecycle using lifecycle- aware components so it starts/stops/cancels automatically; detect app-wide foreground/ background with `ProcessLifecycleOwner` for app-level reactions; and place initialization by how early and how broadly it's needed** — correctness and battery follow from right-scoping.

44.4 High-Level Architecture



Components & ownership

- **Application** does light app-wide init; owns app-scoped singletons (Ch 5).
- `ProcessLifecycleOwner` signals app foreground/background for app-level reactions.
- **Screen lifecycle** scopes visible-only work (`repeatOnLifecycle`); **ViewModel** scopes config-surviving work; **WorkManager** for beyond-UI work.

Direction: process → Application → (app lifecycle via ProcessLifecycleOwner) + (screen lifecycles); work scoped to the matching lifetime.

44.5 Detailed Component Design



Scope	Lifetime	Mechanism	Use for
Process	process	`Application`	app-wide singletons (Ch 5)
App fg/bg	app visible	`ProcessLifecycleOwner`	flush/pause/resume app-wide
Screen	visible	`repeatOnLifecycle(STARTED)` / `lifecycleScope`	UI-only collection
ViewModel	config-change	`viewModelScope` (Ch 6)	survives rotation/nav
Beyond UI	kill/reboot	WorkManager (Ch 30)	guaranteed background

---

44.6 End-to-End Data Flow

App foreground/background reactions



Screen-scoped collection



44.7 Complete Kotlin Implementation

Package: `com.streamcart.app` .

44.7.1 Application: light app-wide init (Ch 45)

```
@HiltAndroidApp
class StreamCartApp : Application(), Configuration.Provider {   // Config.Provider for
WorkManager (Ch 30)
    @Inject lateinit var workerFactory: HiltWorkerFactory
    @Inject lateinit var appLifecycleObserver: AppLifecycleObserver

    override fun onCreate() {
        super.onCreate()
        // Keep this LIGHT (runs on every cold start, Ch 45). Heavy init -> lazy / App Startup.
        Timber.plant(/* redacting tree, Ch 28 */)
        ProcessLifecycleOwner.get().lifecycle.addObserver(appLifecycleObserver)  // app fg/bg
    }
    override val workManagerConfiguration get() =
Configuration.Builder().setWorkerFactory(workerFactory).build()
}
```

44.7.2 App-wide foreground/background observer

```
@Singleton
class AppLifecycleObserver @Inject constructor(
    private val analytics: AnalyticsBatcher,       // Ch 14
    private val realtime: RealtimeConnection,      // Ch 63
    private val sync: SyncScheduler,               // Ch 30
) : DefaultLifecycleObserver {
    override fun onStart(owner: LifecycleOwner) {  // app -> foreground
        realtime.resume()
        sync.requestSync()                          // refresh on return
    }
    override fun onStop(owner: LifecycleOwner) {   // app -> background
        analytics.flush()                           // batch flush (Ch 14)
        realtime.pause()                            // stop expensive work while hidden
    }
}
```

44.7.3 Screen-scoped collection (lifecycle-aware, no leak)

```
// Compose: collect only while visible (Ch 6).
@Composable
fun FeedScreen(viewModel: FeedViewModel = hiltViewModel()) {
    val state by viewModel.uiState.collectAsStateWithLifecycle()   // stops in STOPPED
    // one-time events (Ch 7) with lifecycle-aware collection:
    val lifecycleOwner = LocalLifecycleOwner.current
    LaunchedEffect(viewModel, lifecycleOwner) {
        lifecycleOwner.repeatOnLifecycle(Lifecycle.State.STARTED) {  // visible-only
            viewModel.effects.collect { /* handle */ }
        }
    }
}

// Views: equivalent.
// lifecycleScope.launch { repeatOnLifecycle(Lifecycle.State.STARTED) { vm.state.collect {
render(it) } } }
```

44.7.4 Right-scoping work (the decision)

```
// UI-only (stop when invisible) -> screen scope.
// Survives config change / navigation -> viewModelScope (Ch 6).
class FeedViewModel @Inject constructor(...) : ViewModel() {
    init { viewModelScope.launch { /* work that should survive rotation */ } }
}
// Must survive UI/process (upload, sync) -> WorkManager (Ch 30), NOT a screen/VM scope.
```

44.7.5 Module structure

```
:app   StreamCartApp (@HiltAndroidApp), AppLifecycleObserver
# Screen lifecycle: per feature; WorkManager: :sync (Ch 30); App Startup: Ch 45.
```

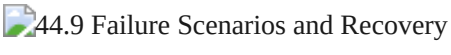
44.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Collect without lifecycle	leak/waste while hidden	`repeatOnLifecycle(STARTED)` / `collectAsStateWithLifecycle`
Long work in screen scope	cancelled on leave	move to `viewModelScope` / WorkManager
Config change restarts work	duplicate	`viewModelScope` survives config change (Ch 6)
Infer bg from one Activity's onStop	false background on rotation/multi-activity	use `ProcessLifecycleOwner`
Callback not unregistered	leak	lifecycle-aware observer auto-removes
Heavy `Application.onCreate`	slow every start	keep light; lazy/App Startup (Ch 45)

- **`ProcessLifecycleOwner` is the correct app-foreground/background signal** — a single Activity's `onStop` fires on rotation and when moving between the app's own Activities, so using it as "backgrounded" is wrong; `ProcessLifecycleOwner` debounces that.
- **Lifecycle-aware components auto-cancel/unregister**, which is the whole leak- prevention story — don't hand-roll `onStart/onStop` registration.

44.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Work runs while backgrounded	battery/network stats	pause on `ON_STOP`; scope correctly	battery saved
Crash collecting after destroy	crash	lifecycle-aware collection	no crash
Leak (Activity retained)	LeakCanary (Ch 89)	remove refs; lifecycle-aware	no leak
Long task dies on navigation	incomplete	WorkManager/ViewModel (Ch 30/6)	survives
False background (rotation)	wrong flush/pause	`ProcessLifecycleOwner` not Activity	correct
Process death	state lost	SavedState/SSOT (Ch 8/3)	restored
Stale data on resume	user report	refresh on `ON_START`	fresh



> **Scope is the decision that prevents most lifecycle bugs:** visible-only → screen; survive-config → ViewModel; survive-process → WorkManager. Choosing wrong causes leaks (too broad), cancelled work (too narrow), or battery drain (unscoped).

44.10 Scaling the Implementation

- **Level 1.** `viewModelScope` + `collectAsStateWithLifecycle` for screens (Ch 6).
- **Level 2.** `ProcessLifecycleOwner` observer for app-wide fg/bg reactions (flush/ pause/resume).
- **Level 3 — production.** Correct scope per work type; light `Application.onCreate`; App Startup for init ordering (Ch 45); leak checks (Ch 89).
- **Level 4 — large.** Modular lifecycle observers; per-feature init on demand (DI, Ch 5); background pausing of expensive subsystems (real-time, Ch 63; location, Ch 67).
- **Level 5 — enterprise.** Lifecycle-driven resource governance, observability of fg/bg time + background work, consistent lifecycle conventions across teams.

Avoid overengineering: most lifecycle needs are met by `viewModelScope` + lifecycle-aware collection + a single `ProcessLifecycleOwner` observer. Don't build a custom lifecycle framework.

44.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Observe lifecycle	Lifecycle-aware (`repeatOnLifecycle`)	manual `onStart/onStop`	Lifecycle-aware always (auto cancel/unregister); manual is leak-prone
App fg/bg	`ProcessLifecycleOwner`	track Activity count	ProcessLifecycleOwner (handles config/multi-activity); manual counting is error-prone
Work scope	Screen / ViewModel / WorkManager	one scope for all	Match scope to needed lifetime
Init	`Application.onCreate` (light) / App Startup	lazy per-feature	App-wide early → Application/App Startup; feature → lazy (Ch 45/5)
Flow collection	`collectAsStateWithLifecycle`	`collectAsState`	Lifecycle-aware default; plain only if always-visible

Lifecycle-aware vs manual callbacks: lifecycle-aware components (`repeatOnLifecycle`, `lifecycleScope`, observers) start/stop/cancel with the lifecycle automatically — no leaked collectors, no work while hidden. Manual `onStart/onStop` registration is verbose and the #1 source of leaks (forgetting to unregister) and crashes (acting on a destroyed view). Always prefer lifecycle-aware.

`ProcessLifecycleOwner` vs Activity-count tracking: both can detect app foreground/background, but `ProcessLifecycleOwner` correctly handles configuration changes and transitions between the app's own Activities (debouncing), whereas manual counting easily miscounts. Use `ProcessLifecycleOwner`.

44.12 Security Considerations

- **Hide sensitive content when backgrounded:** on `ON_STOP` / for sensitive screens, set `FLAG_SECURE` or show a privacy screen so the recents/app-switcher thumbnail and screenshots don't leak content (banking/health).

- **Pause sensitive work in background:** stop location (Ch 67), camera, or real-time streams when the app is hidden unless explicitly a foreground service (Ch 46).
- **Clear sensitive in-memory state on background/process death** where appropriate (e.g., re-auth on return for sensitive apps, Ch 21).
- **Don't log lifecycle with PII** (Ch 28).
- Client vs backend: lifecycle reactions are client resource management; the backend is unaffected but benefits (fewer background requests, cleaner sessions).

44.13 Performance and Resource Optimization

- **Stop work while hidden:** pause polling/real-time/animation on `ON_STOP` — the main battery lever from lifecycle awareness.
- **Light `Application.onCreate`:** it's on the cold-start critical path (Ch 45); defer heavy init (App Startup / lazy).
- **Visible-only collection** (`repeatOnLifecycle`) avoids recomposition/computation for off-screen UI.
- **Release resources** on the right teardown (`onStop` / `onDestroy` / `DisposableEffect`).
- **Batch and flush on background** (analytics, Ch 14) rather than continuously.
- **Profiling:** background CPU/network/wakelocks (Battery Historian), foreground/ background time, LeakCanary (Ch 89), startup trace (Ch 45).

44.14 Testing Strategy

Lifecycle is testable with `TestLifecycleOwner` / `Lifecycle.State` ; ViewModels in isolation (Ch 6).

```
class AppLifecycleObserverTest {
    @Test fun `flushes analytics and pauses on background`() {
        val analytics = FakeAnalytics(); val realtime = FakeRealtime()
        val obs = AppLifecycleObserver(analytics, realtime, FakeSync())
        obs.onStop(TestLifecycleOwner())
        assertTrue(analytics.flushed && realtime.paused)
    }
    @Test fun `reconnects and refreshes on foreground`() {
        val realtime = FakeRealtime(); val sync = FakeSync()
        AppLifecycleObserver(FakeAnalytics(), realtime, sync).onStart(TestLifecycleOwner())
        assertTrue(realtime.resumed && sync.requested)
    }
}

// repeatOnLifecycle behavior: use TestLifecycleOwner + moveToState; assert collection
// starts in STARTED and stops in CREATED.
class ScreenCollectionTest {
    @Test fun `collects only while started`() = runTest {
        val owner = TestLifecycleOwner(initialState = Lifecycle.State.CREATED)
        // launch repeatOnLifecycle(STARTED); move to STARTED -> collects; move to CREATED ->
stops
    }
}
```

What to test: fg/bg observer reactions, lifecycle-scoped collection start/stop (fake lifecycle owner), work-scope correctness (survives config in VM; cancelled in screen), and a leak check (LeakCanary) in instrumentation.

44.15 Production Observability

- **Metrics:** foreground/background time, background CPU/network/wakelocks, cold-start frequency (Ch 45), leak reports (Ch 89), analytics-flush-on-background success.
- **Logs:** `app_lifecycle{event=foreground/background}`, `scope{work, lifecycle}` (no PII, Ch 28).
- **Debugging:** battery drain → work not paused in background / wrong scope; crashes after navigation → non-lifecycle-aware collection; leaks → retained Activity/ callback; stale on resume → no foreground refresh; slow launch → heavy `Application.onCreate` (Ch 45).
- **Alerting:** background-work/wakelock spike → lifecycle regression.
- **Privacy:** sensitive content in recents → `FLAG_SECURE`/privacy screen.

44.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Lifecycle hierarchy?* Process → Application → Activity/Fragment/Compose, plus app-wide foreground/background.
2. *What runs once per process?* `Application.onCreate` (every cold start).
3. *What survives config change?* ViewModel (Ch 6); not the Activity.
4. *What survives process death?* Only persisted state (SavedState/SSOT, Ch 8/3).
5. *Lifecycle-aware components?* `LifecycleOwner`/observer, `repeatOnLifecycle`, `lifecycleScope`, `viewModelScope` — auto start/stop/cancel.

App fg/bg 6. *Detect app foreground/background?* `ProcessLifecycleOwner` (ON_START/ON_STOP). 7. *Why not a single Activity's onStop?* Fires on rotation/multi-activity → false background. 8. *What to do on background?* Flush analytics (Ch 14), pause real-time (Ch 63), stop expensive work. 9. *On foreground?* Reconnect/refresh.

Scoping 10. *Visible-only work?* Screen scope (`repeatOnLifecycle(STARTED)`). 11. *Survive config/nav?* `viewModelScope`. 12. *Survive UI/process?* WorkManager (Ch 30). 13. *Classic scoping bug?* Long work in a screen scope (cancelled on leave).

Leaks 14. *How avoid leaks?* Lifecycle-aware observation; no Activity refs; auto-unregister. 15. *`collectAsState` vs `collectAsStateWithLifecycle`?* The latter stops in STOPPED. 16. *Compose lifecycle?* `LocalLifecycleOwner`, `DisposableEffect`/`LaunchedEffect`.

Init 17. *Where app-wide init?* `Application.onCreate` (light) or App Startup (Ch 45). 18. *Why keep onCreate light?* Cold-start critical path. 19. *Per-feature init?* Lazy via DI (Ch 5).

Security 20. *Sensitive content in recents?* `FLAG_SECURE`/privacy screen on background. 21. *Pause sensitive work?* Location/camera/streams when hidden.

Testing 22. *How test fg/bg?* Fake observer + onStart/onStop; assert reactions. 23. *How test lifecycle collection?* `TestLifecycleOwner` + move states.

Production debugging 24. *Battery drain?* Work not paused/wrong scope. 25. *Crash after leaving screen?* Non-lifecycle-aware collection. 26. *Stale on resume?* No foreground refresh.

Design-change challenges

1. **Flush analytics when the app backgrounds.** `ProcessLifecycleOwner` `onStop`.
2. **Stop a WebSocket while hidden.** Pause on background, reconnect on foreground.
3. **A long task dies on navigation.** Move to WorkManager/ViewModel.
4. **Battery drain from background polling.** Pause on background; scope correctly.
5. **Sensitive content shows in recents.** `FLAG_SECURE`/privacy screen.

Mock walkthrough (abridged)

Clarify (what work, how long should it live, app-wide reactions?) → lifecycle hierarchy → scope work (screen/VM/WorkManager) with lifecycle-aware components → `ProcessLifecycleOwner` for app fg/bg → light init placement (Ch 45) → leak avoidance → security (FLAG_SECURE) → testing.

44.17 Practical Exercises

1. **Beginner.** Add a `ProcessLifecycleOwner` observer that flushes analytics on background and refreshes on foreground. *Eval:* correct reactions.
2. **Intermediate.** Make a screen collect its Flow only while visible (`repeatOnLifecycle` / `collectAsStateWithLifecycle`). *Eval:* stops when hidden; no leak.
3. **Advanced.** Right-scope three work items (visible-only, survive-config, survive-process) to screen/VM/WorkManager. *Eval:* each has correct lifetime.
4. **Production debugging.** Fix battery drain (background polling not paused) and a crash (collecting after destroy). *Eval:* paused; lifecycle-aware.
5. **System design challenge.** Add `FLAG_SECURE`/privacy screen on background for a sensitive screen and re-auth on foreground (Ch 21). *Eval:* no recents leak; gated return.

(Solutions withheld unless requested.)

44.18 Chapter Summary & Reference Sheet

Key decisions: understand the **lifecycle hierarchy** (process → Application → screen, plus app foreground/background); **scope every piece of work to the lifetime it needs** — visible-only → **screen scope** (`repeatOnLifecycle` / `collectAsStateWithLifecycle`), survive-config/nav → **`viewModelScope`** (Ch 6), survive-UI/process → **WorkManager** (Ch 30) — using **lifecycle-aware components** so it auto-cancels (no leaks/waste); detect **app foreground/background** with **`ProcessLifecycleOwner`** (not a single Activity's `onStop`) for app-level reactions (flush analytics Ch 14, pause real-time Ch 63, refresh on return); keep **`Application.onCreate` light** (Ch 45); and **protect sensitive content** on background (`FLAG_SECURE`).

Concepts: lifecycle hierarchy, lifecycle-aware components, `ProcessLifecycleOwner`, scope selection (screen/ViewModel/WorkManager), app fg/bg reactions, init placement, leak avoidance, Compose lifecycle.

Common mistakes: long work in a screen scope (cancelled); unscoped background work (battery); non-lifecycle-aware collection (leak/crash); inferring background from one Activity's `onStop`; heavy `Application.onCreate`; retained Activity refs; sensitive content in recents.

Checklist: ☐ scope work correctly (screen/VM/WorkManager) ☐ lifecycle-aware collection ☐ `ProcessLifecycleOwner` for app fg/bg ☐ flush/pause on background + refresh on foreground ☐ light `Application.onCreate` ☐ no retained Activity refs ☐ `FLAG_SECURE`/privacy on background (sensitive) ☐ release resources on teardown.

Key APIs: `Application` / `@HiltAndroidApp`, `ProcessLifecycleOwner`, `DefaultLifecycleObserver`, `Lifecycle.State` / `repeatOnLifecycle`, `lifecycleScope` / `viewModelScope`, `collectAsStateWithLifecycle`, `LocalLifecycleOwner`, `DisposableEffect` / `LaunchedEffect`, `FLAG_SECURE`, `TestLifecycleOwner`.

References:

- Lifecycle-aware components: https://developer.android.com/topic/libraries/architecture/lifecycle
- Process & app lifecycle: https://developer.android.com/guide/components/activities/process-lifecycle
- ProcessLifecycleOwner: https://developer.android.com/reference/androidx/lifecycle/ProcessLifecycleOwner

- Safer flow collection: https://developer.android.com/topic/libraries/architecture/coroutines#restart
- (composes) Ch 6 UI State, Ch 8 Process Death, Ch 30 WorkManager, Ch 45 Startup

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; Application init + ProcessLifecycleOwner observer + lifecycle-aware collection + scope decision; concurrency (ProcessLifecycleOwner vs Activity, auto-unregister); failure (battery/leak/scope); 5 levels; lifecycle-aware-vs-manual trade-offs; security (FLAG_SECURE); perf (pause-while-hidden); fake-lifecycle tests; observability (fg/bg time); 26 Q&A + 5 challenges. ✅
- **Expand later:** startup (Ch 45), services/Doze (Ch 46/47), process death (Ch 8), real-time pause (Ch 63).
- **Interview coverage:** exceeds ≥25 bar.